Hands-On Activity: Explore movie data with pivot tables

Activity overview

In the previous video, you were introduced to pivot tables as a tool for quickly comparing metrics, performing calculations, and generating readable reports. In this activity, you'll navigate an example of a workplace scenario where you will create a pivot table to answer stakeholder questions and visualize data.

By the time you complete this activity, you will be able to apply pivot tables in your own analysis projects. This will enable you to draw insights and create reports directly from your spreadsheets, which is important for your career as a data analyst.

Review the following scenario. Then complete the step-by-step instructions.

You are a data analyst working with a filmmaking company. The company is trying to identify what genre of film they should make next. In order to help them make this decision, your manager has asked you to answer the following questions:

- On average, how much money do movies make in each genre?
- On average, how much money is spent on a movie broken by genre?
- On average, which genre has the most profitable movies?

Pivot tables make answering these types of questions much quicker and more accurate when compared to doing these same calculations by hand!

Follow the instructions to complete each step of the activity. Then answer the questions at the end of the activity before going to the next course item.

To get started, access the movie spreadsheet from the previous video.

Select the link to the movie spreadsheet to create a copy. If you don’t have a Google account, you may download the data directly from the attachments below.

Link to movie data: [Movie Data Starter Project]

OR

For this exercise, you will use Google Sheets to work with pivot tables. If you are using Excel, the interface might be different. However, the majority of the functions are similar.

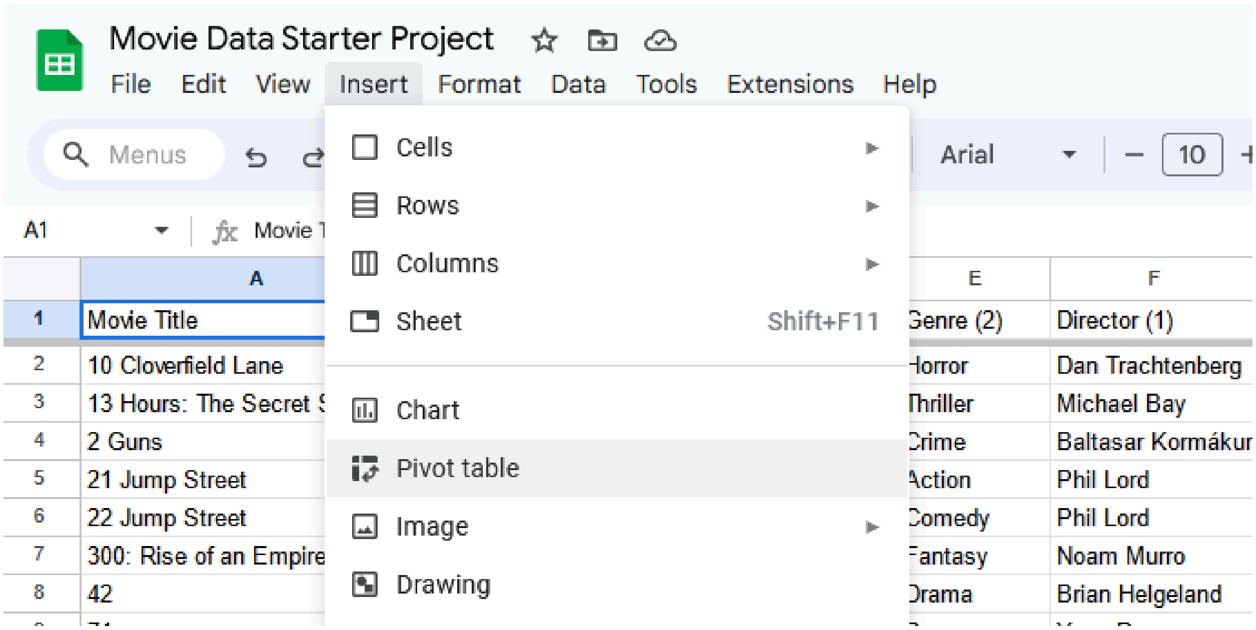
First, you need to generate your pivot table:

1. Open the Movie Data Starter Project spreadsheet.

2. Navigate to the **Movie Data** sheet.

3. With your cursor in a non-blank field, select the **Insert** menu.

4. Select **Pivot Table**.



5. Verify the following **Create pivot table** settings:

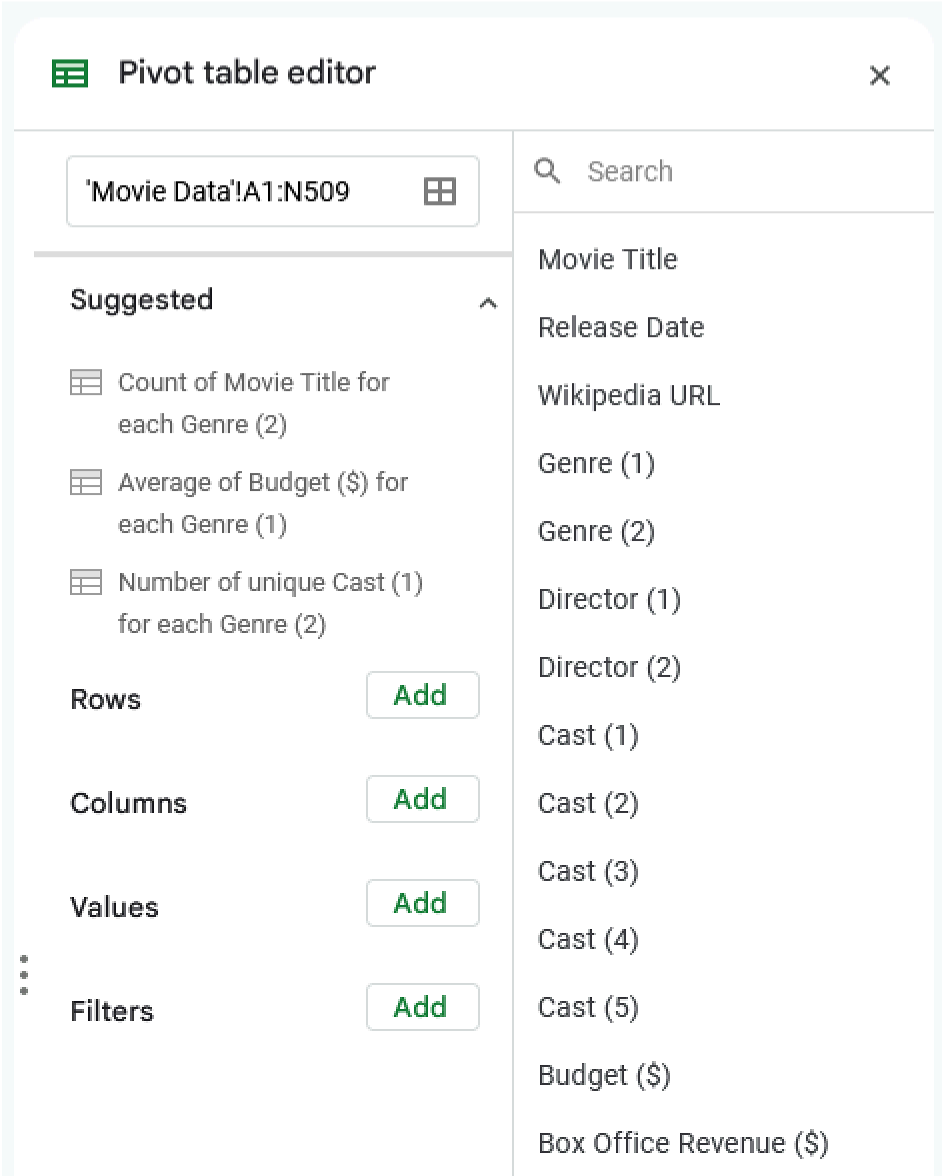
a. Data range = `'Movie Data'!A1:N509`

b. Insert to = `New Sheet`

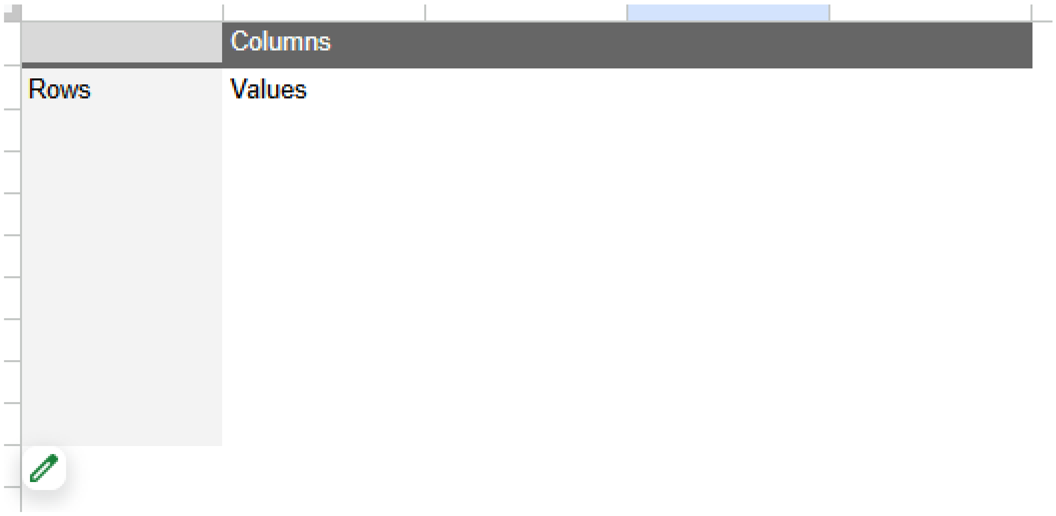
6. Select **Create**.

7. In the newly created sheet, right-select on the tab and select **Rename** to rename the sheet to Summary.

8. Now, your Pivot table editor is likely going to be open on the right side of the screen.



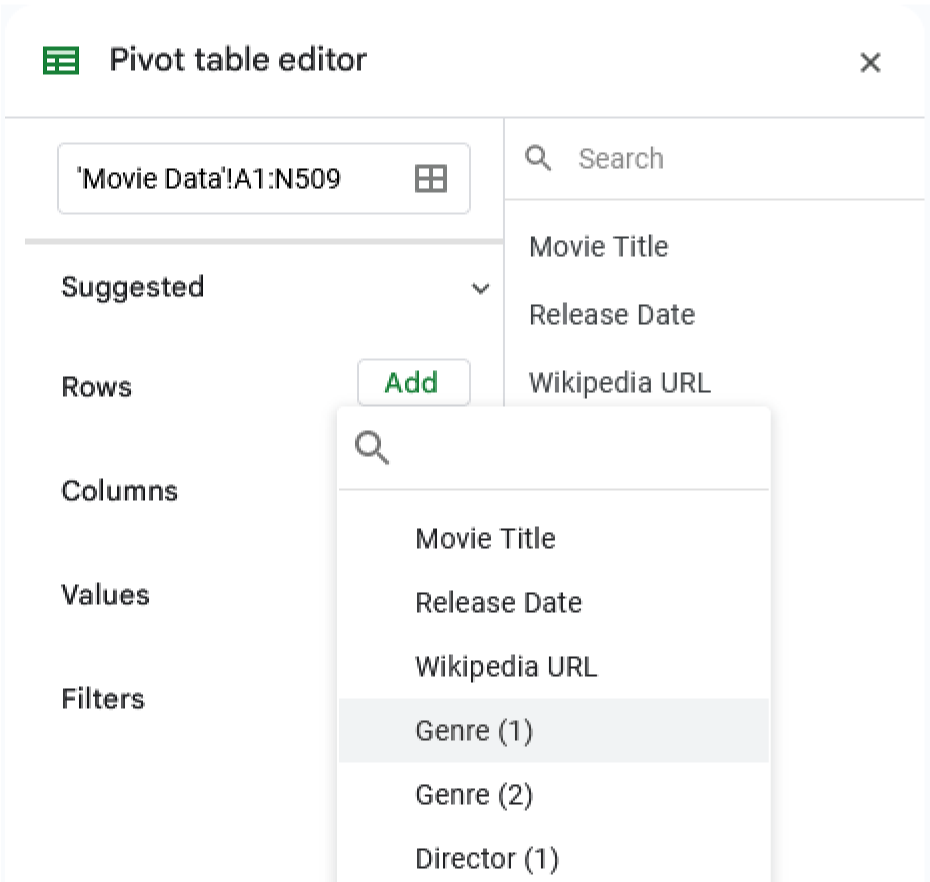
If the Pivot table editor isn't already open, move your cursor over the pivot table area to reveal a small green pencil **Edit** icon. Select it to open the **Pivot table editor**.

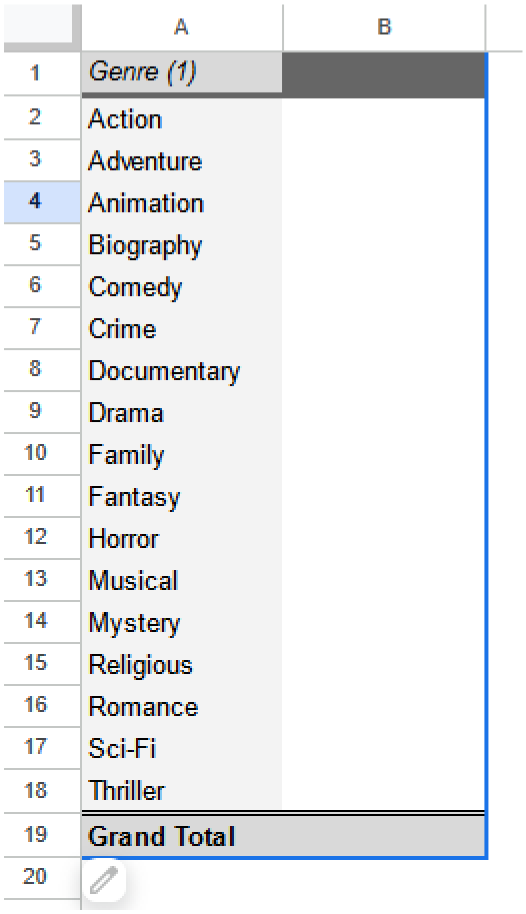


In the Pivot table editor:

1. Select on the **Add** button to the right of the **Rows** section.

2. Select **Genre (1)**.


Pivot table editor
'Movie Data'!A1:N509
Search
Suggested
Rows
Add
Columns
Values
Filters
Movie Title
Release Date
Wikipedia URL
Movie Title
Release Date
Wikipedia URL
Genre (1)
Genre (2)
Director (1)



3. Next, select on the **Add** button to the right of the Values section.

4. Select **Box Office Revenue ($)**.

5. In that smaller box, change the **Summarize by** selection by using the dropdown and changing the summarization from **SUM** to **AVERAGE**.

Genre (1)	AVERAGE of Bo
Action	233,839,500.00
Adventure	308,633,333.33
Animation	276,200,000.00
Biography	58,806,666.67
Comedy	123,442,857.14
Crime	57,669,565.22
Documentary	68,500,000.00
Drama	80,990,337.08
Family	270,958,333.33
Fantasy	244,610,000.00
Horror	75,646,511.63
Musical	130,000,000.00
Mystery	96,780,000.00
Religious	36,914,285.71
Romance	53,705,384.62
Sci-Fi	255,443,571.43
Thriller	136,937,500.00
Grand Total	**151,983,208.66**

Now you have the breakdown of Average Box Office Revenue ($) by Genre.

Now, investigate the budget of these movies so you can compare them to the revenue you just calculated and determine their profit. Repeat the previous process to find the Budget ($):

1. Select on the **Add** button to the right of the **Values** section.

2. Select **Budget ($)**.

3. In that smaller box, change the **Summarize by** selection by using the dropdown and changing the summarization from **SUM** to **AVERAGE**.

Genre (1)	AVERAGE of Bo	AVERAGE of Bu
Action	233,839,500.00	82,810,000.00
Adventure	308,633,333.33	82,828,571.43
Animation	276,200,000.00	89,666,666.67
Biography	58,806,666.67	28,760,000.00
Comedy	123,442,857.14	39,914,285.71
Crime	57,669,565.22	30,443,478.26
Documentary	68,500,000.00	10,000,000.00
Drama	80,990,337.08	26,785,955.06
Family	270,958,333.33	78,483,333.33
Fantasy	244,610,000.00	91,000,000.00
Horror	75,646,511.63	14,688,372.09
Musical	130,000,000.00	40,250,000.00
Mystery	96,780,000.00	22,600,000.00
Religious	36,914,285.71	10,142,857.14
Romance	53,705,384.62	19,730,769.23
Sci-Fi	255,443,571.43	65,254,285.71
Thriller	136,937,500.00	33,243,750.00
Grand Total	**151,983,208.66**	**48,871,397.64**

Now you have the average revenue and budget broken down by genre.

Note: You may not be able to complete this part of the activity in Microsoft Excel. If you're completing this activity in Microsoft Excel and do not find a **Custom** option in step 5.4, move on to step 6, complete the activity in Google Sheets, or use the resources provided earlier in this lesson to complete this step.

You can also use pivot tables to calculate profitability directly in the pivot table:

1. Select the **Add** button to the right of the **Values** section.

2. Select **Calculated Field**.

3. Copy and paste the following into the **formula**:

```
=AVERAGE('Box Office Revenue ($)')-AVERAGE('Budget ($)')
```

4. Change the **Summarize by** selection from **SUM** to **Custom**.

5. Select on cell **D1** (in your Summary tab) and enter **Average Profit** to rename the newly calculated field.

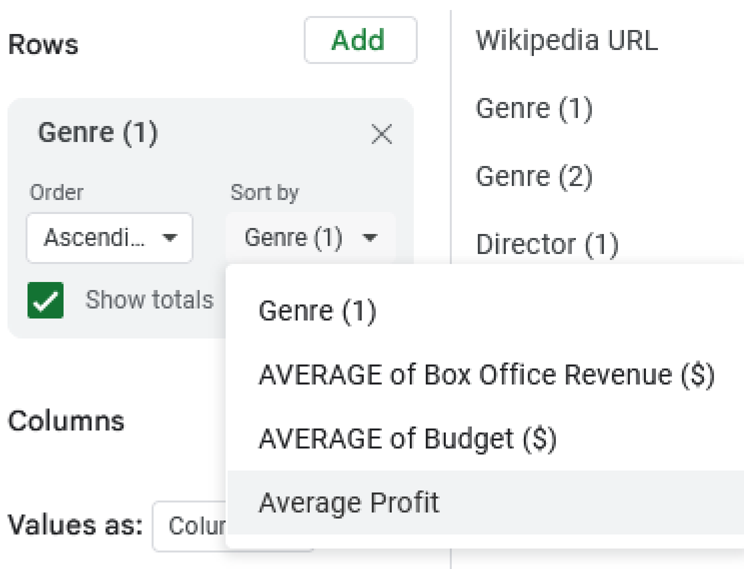
Genre (1)	AVERAGE of Box Office Revenue ($)	AVERAGE of Budget ($)	Average Profit
Action	233,839,500.00	82,810,000.00	151,029,500.00
Adventure	308,633,333.33	82,828,571.43	225,804,761.90
Animation	276,200,000.00	89,666,666.67	186,533,333.33
Biography	58,806,666.67	28,760,000.00	30,046,666.67
Comedy	123,442,857.14	39,914,285.71	83,528,571.43
Crime	57,669,565.22	30,443,478.26	27,226,086.96
Documentary	68,500,000.00	10,000,000.00	58,500,000.00
Drama	80,990,337.08	26,785,955.06	54,204,382.02
Family	270,958,333.33	78,483,333.33	192,475,000.00
Fantasy	244,610,000.00	91,000,000.00	153,610,000.00
Horror	75,646,511.63	14,688,372.09	60,958,139.53
Musical	130,000,000.00	40,250,000.00	89,750,000.00
Mystery	96,780,000.00	22,600,000.00	74,180,000.00
Religious	36,914,285.71	10,142,857.14	26,771,428.57
Romance	53,705,384.62	19,730,769.23	33,974,615.38
Sci-Fi	255,443,571.43	65,254,285.71	190,189,285.71
Thriller	136,937,500.00	33,243,750.00	103,693,750.00
Grand Total	**151,983,208.66**	**48,871,397.64**	**103,111,811.02**

Perhaps your manager also asks you if there's any way to sort the sheet to make it easier to identify which genres are most profitable. To do that, follow these steps:

6. In the Pivot table editor, go to the **Genre (1)** in your rows and update these settings:

a. For the **Sort by** dropdown, change the value from **Genre (1)** to **Average Profit**.

b. For the **Order** dropdown, change the value from **Ascending** to **Descending**.



Genre (1)	AVERAGE of Box Office Revenue ($)	AVERAGE of Budget ($)	Average Profit
Adventure	308,633,333.33	82,828,571.43	225,804,761.90
Family	270,958,333.33	78,483,333.33	192,475,000.00
Sci-Fi	255,443,571.43	65,254,285.71	190,189,285.71
Animation	276,200,000.00	89,666,666.67	186,533,333.33
Fantasy	244,610,000.00	91,000,000.00	153,610,000.00
Action	233,839,500.00	82,810,000.00	151,029,500.00
Thriller	136,937,500.00	33,243,750.00	103,693,750.00
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Horror	75,646,511.63	14,688,372.09	60,958,139.53
Documentary	68,500,000.00	10,000,000.00	58,500,000.00
Drama	80,990,337.08	26,785,955.06	54,204,382.02
Romance	53,705,384.62	19,730,769.23	33,974,615.38
Biography	58,806,666.67	28,760,000.00	30,046,666.67
Crime	57,669,565.22	30,443,478.26	27,226,086.96
Religious	36,914,285.71	10,142,857.14	26,771,428.57
Grand Total	**151,983,208.66**	**48,871,397.64**	**103,111,811.02**

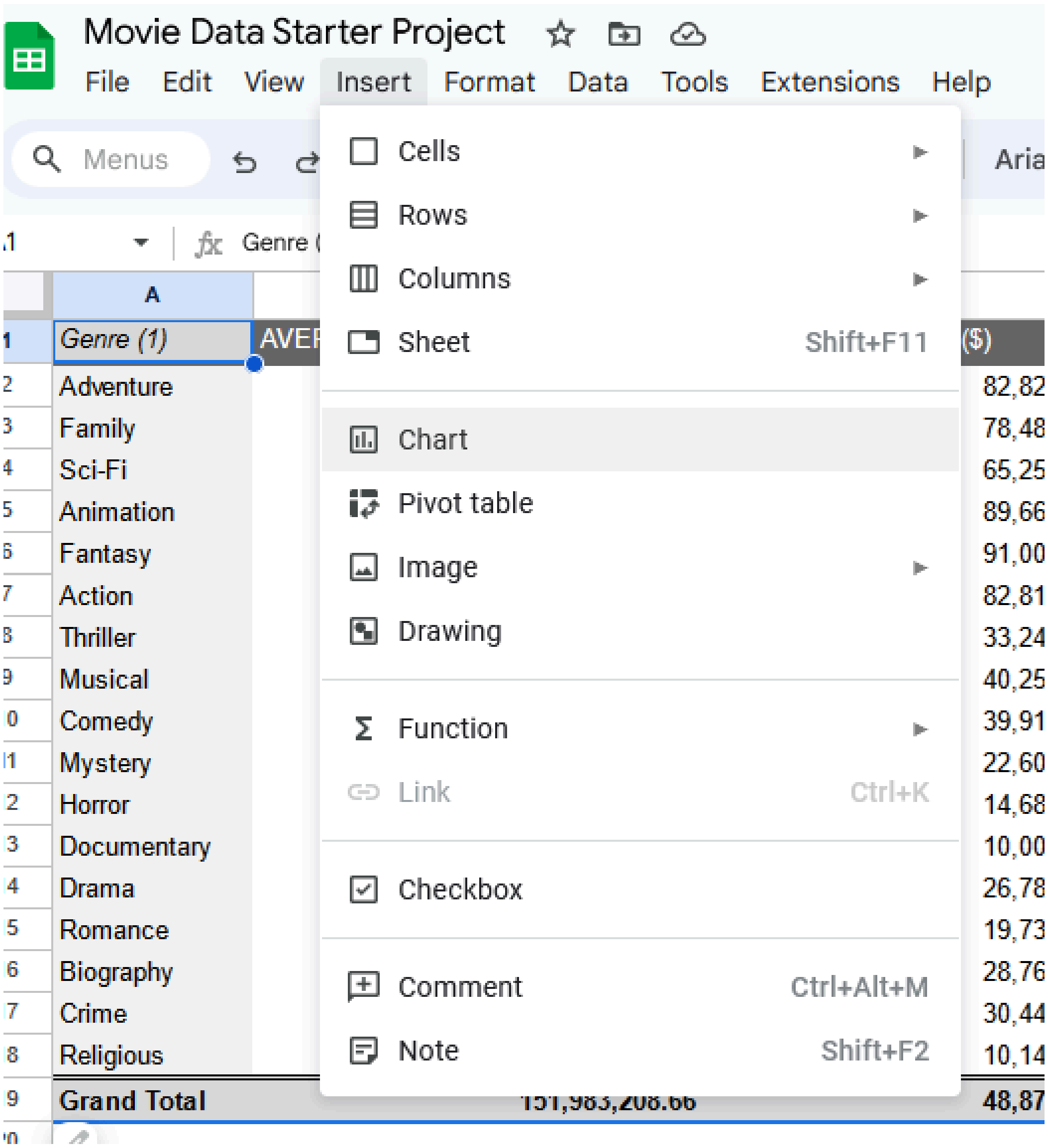
Here is that pivot table, now sorted to have the most profitable genres as the top of the list!

Tables provide useful information, but visualizations can be even more effective for communicating data insights.

1. Select any cell in your pivot table.

2. Navigate to the **Insert** menu.

3. Select **Chart**.



This creates a chart in the same worksheet as your pivot table.

4. Now, move the chart to the side of your table:

a. Select the **chart**.

b. Left-select in the chart and—while holding down your cursor—drag the **cursor** next to your table. This should move the chart with the movement of your cursor.

5. Next, set up your chart.

a. In the chart editor, navigate to **Setup**.

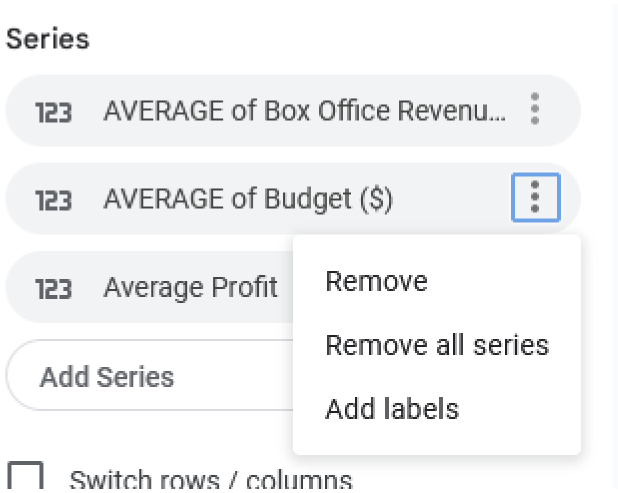
b. In the **Setup** window, change the **Chart Type** to **Bar chart**.

c. Under **Series**, select on the three dots to the right of **AVERAGE of Budget ($)**.

d. Select **Remove**.

e. Next, under **Series**, select the three dots to the right of **AVERAGE of Profit**.

f. Select **Remove**.

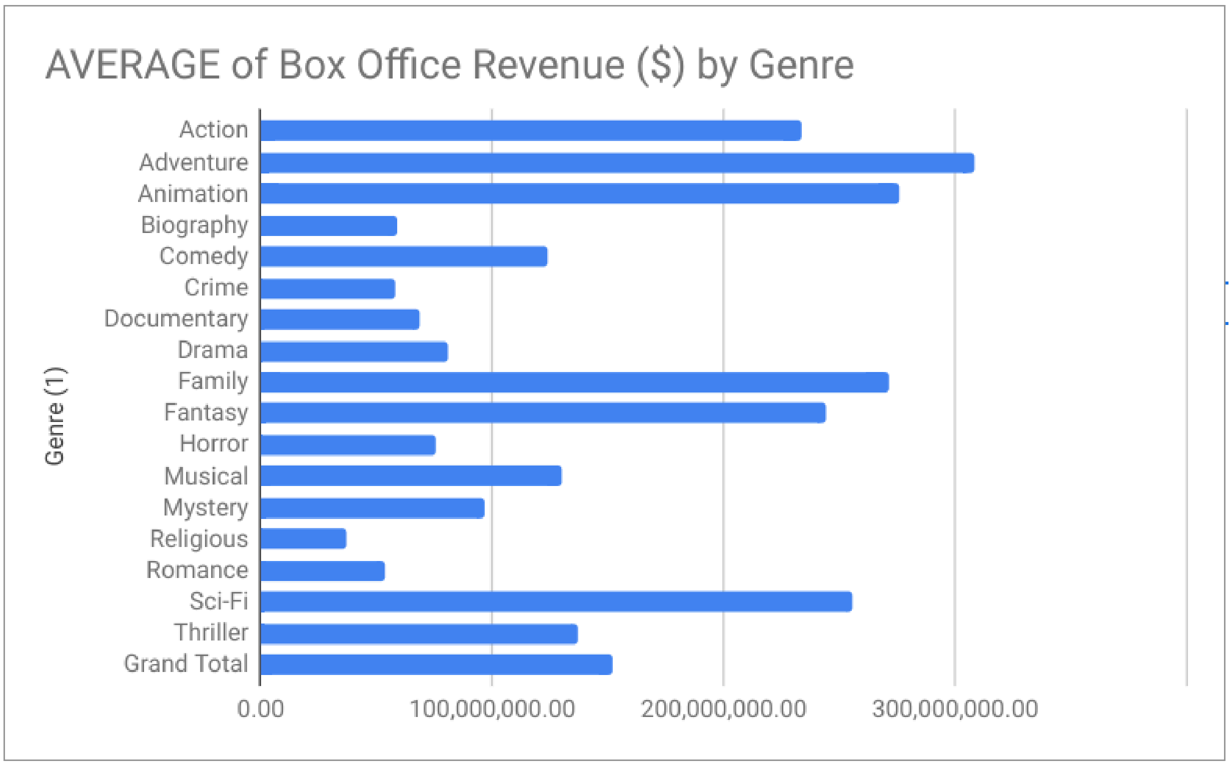


6. Next, customize your chart.

a. In the chart editor, navigate to **Customize**.

b. Under **Chart & Axis Titles**, select **Chart title**.

c. Change the chart title to **Average of Box Office Revenue ($) by Genre**.



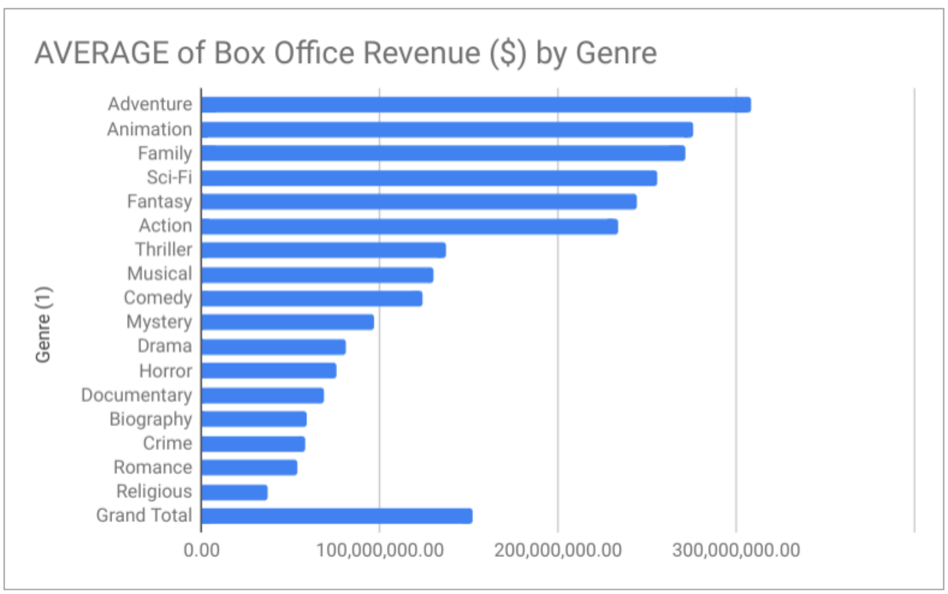
Great work! Now you have a visual of box office revenue by genre.

7. The graph you created displays genre in alphabetical order. However, if you're interested in knowing which genre garners the highest average box office revenue ($), you'll want to sort the data from highest to lowest. You can do that by sorting the information in the pivot table. To do this:

a. Open the Pivot table editor.

b. In the Pivot table editor, go to the **Genre (1)** in your rows and update these settings:

1. For the **Sort by** dropdown, select **Average of Box Office Revenue ($)**.
2. For the **Order** dropdown, select **Descending**.



Note: You may have noticed that there may be a **Grand Total**, which doesn't appear to fall into the middle of your sorting. That's okay! This is because it's the total across all genres. You can choose to remove this by going back into your pivot table editor, in your Rows section, uncheck the **Show totals** box. This should automatically update any other references (charts or other) that were pointed at your pivot table!

Be sure to save a copy of the spreadsheet template you used to complete this activity. You can use it for further practice or to help you work through your thought processes for similar tasks in a future data analyst role.

Using pivot tables, you were able to answer all three of your stakeholders' questions about movie revenue, budget, and profit by genre! Not only that, you presented your findings in an organized way to make your findings clearer to your audience and provided them with a useful visualization.

Pivot tables are incredibly powerful. The ability to calculate these types of values will allow you to be more efficient in how you work, as well.

1. Adjust your pivot table to include information about directors [use the **Director (1)** field] instead of genre. Which director had the highest average box office revenue? 1 / 1 point

- ○ Tim Miller
- ○ Michael Mann
- ◉ Peter Jackson
- ○ Zach Snyder

Correct
Peter Jackson averaged $956M in box office revenue, which is the highest box office revenue in this dataset.

2. In the text box below, write 2-3 sentences (40-60 words) in response to each of the following questions: 1 / 1 point

- How can using pivot tables directly in a spreadsheet help you analyze data in the future?
- What are some of the benefits of being able to summarize data directly in your spreadsheet?

Answer: If there are more than 20 columns and the dataset is huge, and we want to analyze 2 to 3 columns, then a pivot table is the best option. Using a pivot table, we can analyze thoroughly and also get a visualization of the particular columns.

Benefits of the pivot table are,

We can choose the particular column and analyze it.

We can also use different types of operations as per requirements.

We can also use different types of charts as per requirements.

We can analyze and visualize particular portion of the huge data as per requirements, etc.

Feedback

Congratulations on completing this hands-on activity! In this activity you created a pivot table and some basic visualizations directly in your spreadsheet to gain insight into your data. An effective response would include that this will allow you to analyze data quickly using one analysis tool. This can help you quickly find answers for stakeholders and generate shareable reports to share your findings.